PAPER CODE NO.
COMP 575

EXAMINER: Prof. J. Goulermas
DEPARTMENT: Computer Science

Second Semester Examinations 2018-19

Computational Intelligence

TIME ALLOWED: THREE HOURS

INSTRUCTIONS TO CANDIDATES

The numbers in the right hand margin represent a guide to the marks available for that question (or part of a question). Total marks available are 100.

You must answer ALL FOUR questions.

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1. a) Design a single-layer perceptron with two inputs denoted by x_1 and x_2, and one threshold corresponding to a fixed input denoted as $x_0 = -1$. The perceptron is based on a signum activation function ϕ, of output values +1 (for non-negative input) and −1 (for negative input). Starting with the initial weights w_1= -0.4, w_2= +0.8 and an initial threshold w_0= +0.3, calculate all the steps and weight adjustments involved in its training, using the perceptron training algorithm and a learning rate η= 0.5. The goal is to teach the network to learn the two-input OR problem. **10**

   Hint: During training, in each epoch, present the input patterns in the following order: (0,0), (0,1), (1,0) and (1,1). Also, map the "false" (or 0) output of OR to class label −1 and the "true" to +1.

   b) In the context of multi-layer feed-forward neural networks:

      i) Use diagrams and equations to describe in detail the forward and backward passes of the back-propagation learning algorithm (you do not need to derive the equations, but use references to link your equations to the neuron and weight indices within your diagrams). **8**

      ii) Give two examples of suitable activation functions and describe their main characteristics. **3**

   c) Describe the basic modelling characteristics of Boltzmann learning and its weight adaptation rule. **4**

Total 25

2. a) With regard to the support vector machines classifier, describe how it models linear decision boundaries, and derive the associated equations of the separation margin and the support vectors. **8**

b) For the case of single-layer perceptron neural networks with linear activation functions, derive the linear least squares training algorithm used for regression when all input patterns and desired outputs are available at once. **7**

c) Design a mechanism by which the weights of radial basis function neural networks can be adapted, given a set of training patterns, the centres and the width parameters of their basis functions. **6**

d) Describe the main characteristics of supervised and unsupervised learning in neural networks and machine learning. **4**

Total 25

3. a) Describe three different crossover schemes that are suitable to a genetic algorithm with real-valued representation (that is, one whose chromosomes are vectors of real numbers). **5**

b) Why do genetic algorithms need to handle invalid chromosomes? Describe three different ways for handling such cases. **4**

c) Design a mechanism of your choice which would enable the mutation rate to adapt to parental or population diversity. State whether this mechanism will work for a discrete or a real-valued genotype representation. **5**

d) Explain the mechanisms of uniform crossover and binomial crossover in discrete chromosome representations. **3**

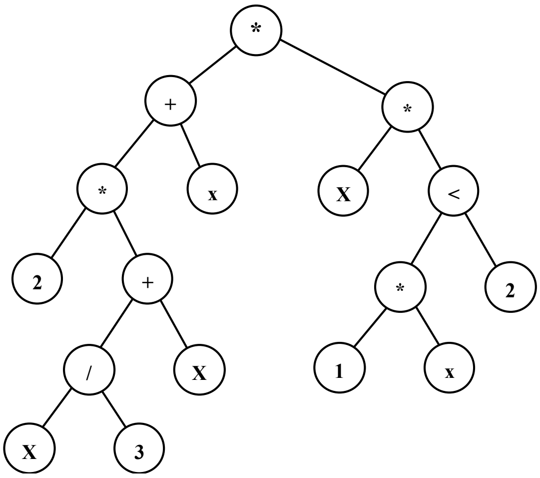
e) Analyse how the number of crossover points affects the positional bias, the distributional bias and the recombination power of the crossover. **4**

f) What is the population model with niching, how is it modelled, and what is its primary advantage? **4**

Total 25

4. a) What is the basic architecture of genetic programming, and how are its crossover and mutation operators typically defined to operate? **5**

b) In the context of genetic programming, describe what is meant by an intron, and give an example with the aid of the tree in the figure below: **4**



c) Describe what is gene expression programming, and provide examples for the functionalities of its crossover and mutation operators. **5**

d) How is the particle swarm optimisation modelled, and what are the basic equations for updating its particle positions and velocities? **5**

e) Design one particle swarm optimisation variant to the basic method and clarify its advantages and the way it differs from the standard model. **2**

f) Provide the basic update equations and encoding scheme for the bio-inspired type of optimisation techniques referred to as evolutionary strategies. **4**

Total 25

PAPER CODE NO.	EXAMINER:	Dr Goulermas
COMP 575	DEPARTMENT:	Computer Science

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INSTRUCTIONS TO CANDIDATES

The numbers in the right hand margin represent a guide to the marks available for that question (or part of a question). Total marks available are 100.

You must answer ALL FOUR questions.

1. a) Describe three possible mechanisms that can be used to set the centre and width parameters of Gaussian activation functions in radial basis function (RBF) neural networks. **6**

b) For the case of single-layer perceptron (SLP) neural networks with linear activation functions, derive the linear least squares (LLS) training algorithm, used when all input patterns and desired outputs are available at once. **7**

c) For the self organising maps (SOMs), describe the main elements of the competitive, the cooperative and the adaptive processes which enable such networks to learn. **8**

d) Describe the main characteristics of supervised and unsupervised learning in neural networks and machine learning. **4**

Total 25

2. a) Design a single-layer perceptron with two inputs denoted by x_1 and x_2, and one threshold corresponding to a fixed input denoted as $x_0 = -1$. The perceptron is based on a signum activation function ϕ, of output values +1 (for non-negative input) and −1 (for negative input). Starting with the initial weights $w_1 = +0.5$, $w_2 = -1.3$ and an initial threshold $w_0 = -2.2$, calculate all the steps and weight adjustments involved in its training, using the perceptron training algorithm and a learning rate $\eta=0.5$. The goal is to teach the network to learn the two-input AND problem. **10**

Hint: During training, in each epoch, present the input patterns in the following order: (0,0), (0,1), (1,0) and (1,1). Also, map the "false" (or 0) output of AND to class label −1 and the "true" to +1.

b) Describe the concepts and derive the associated equations of the separation margin and the support vectors, used in support vector machines (SVMs). **8**

c) For the case of multi-layer perceptron (MLP) neural networks, use diagrams and equations to describe the forward and backward passes of the back-propagation learning algorithm (you do not need to derive the equations). **7**

Total 25

3. a) Why do genetic algorithms need to handle invalid chromosomes? Describe three different ways for handling such cases. **4**

   b) How does Baker's linear ranking work and how can it avoid the close-race and the super-individual cases? **4**

   c) Explain why parent selection is needed in genetic algorithms and provide three such selection schemes. **4**

   d) For string shaped chromosomes with discrete representations, describe how the number of crossover points affects the positional bias, the distributional bias and the recombination power of the crossover. **4**

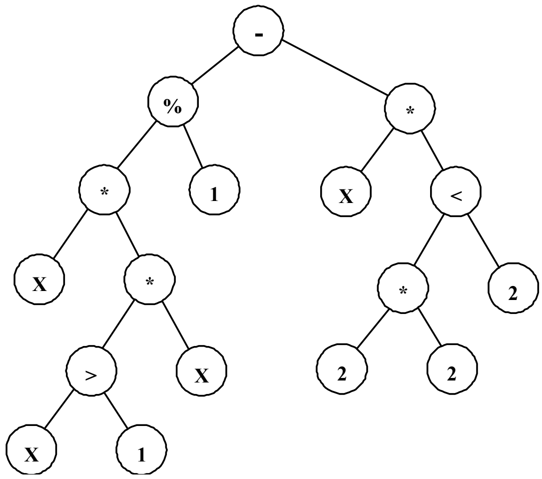
   e) Design a mechanism of your choice which would enable the mutation rate to adapt to parental or population diversity. State whether this mechanism will work for a discrete or real-valued genotype representation. **5**

   f) Describe two different crossover schemes that are applicable to a genetic algorithm that employs chromosomes based on a real-valued representation. **4**

Total 25

4. a) What is the basic architecture of genetic programming, and how are its crossover and mutation operators typically defined to operate? **5**

b) In the context of genetic programming, describe what is meant by an intron, and give an example with the aid of the tree in the figure below: **4**



c) Describe what is gene expression programming, and provide examples for the functionalities of its crossover and mutation operators. **5**

d) Describe two particle swarm optimisation variants, and clarify their advantages as well as how they differ from the standard model. **5**

e) For the type of optimisation techniques referred to as evolutionary strategies:

i) Explain their basic update equations and encoding scheme. **4**

ii) Discuss one possible mechanism that extends their basic model. **2**

Total 25